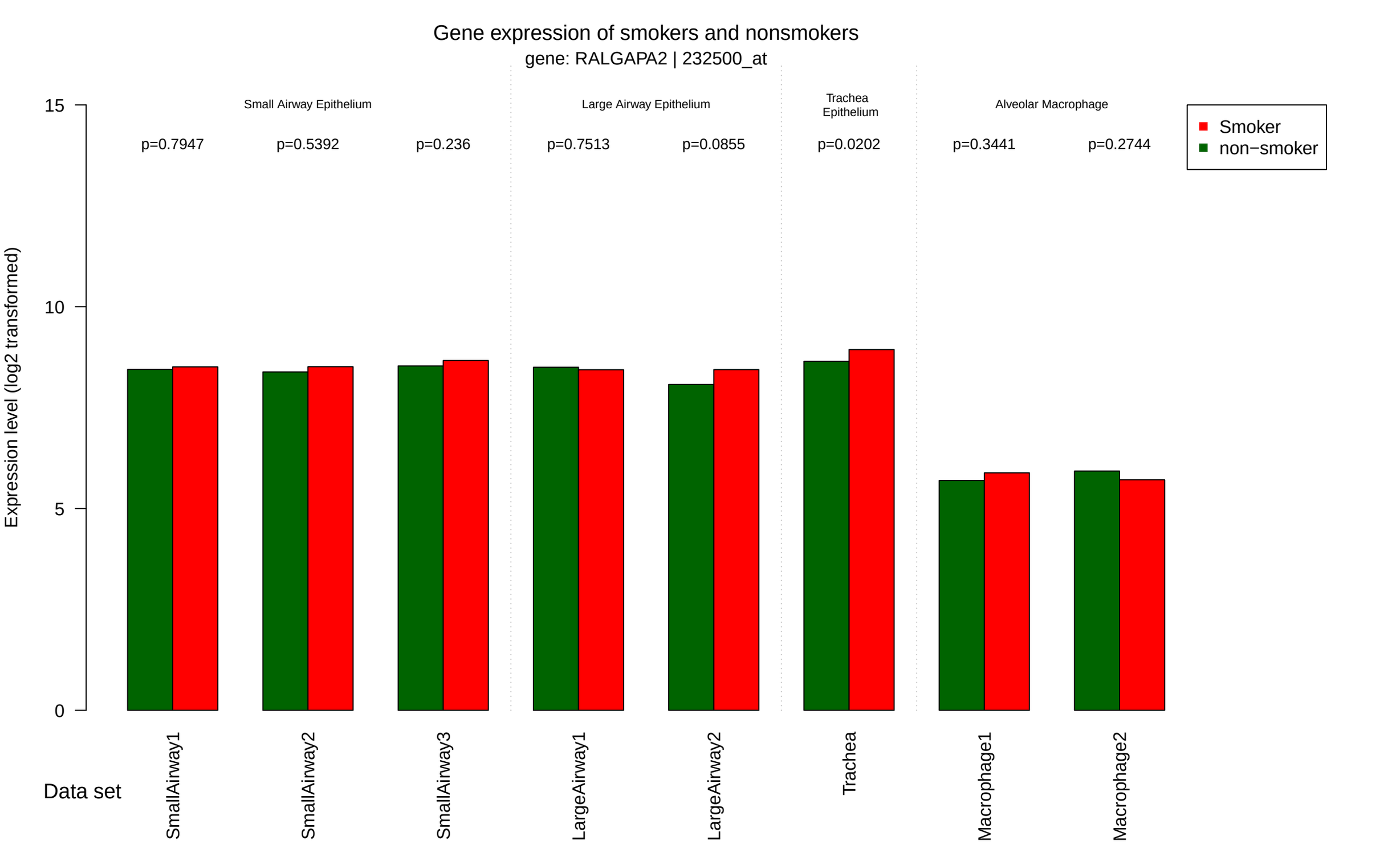
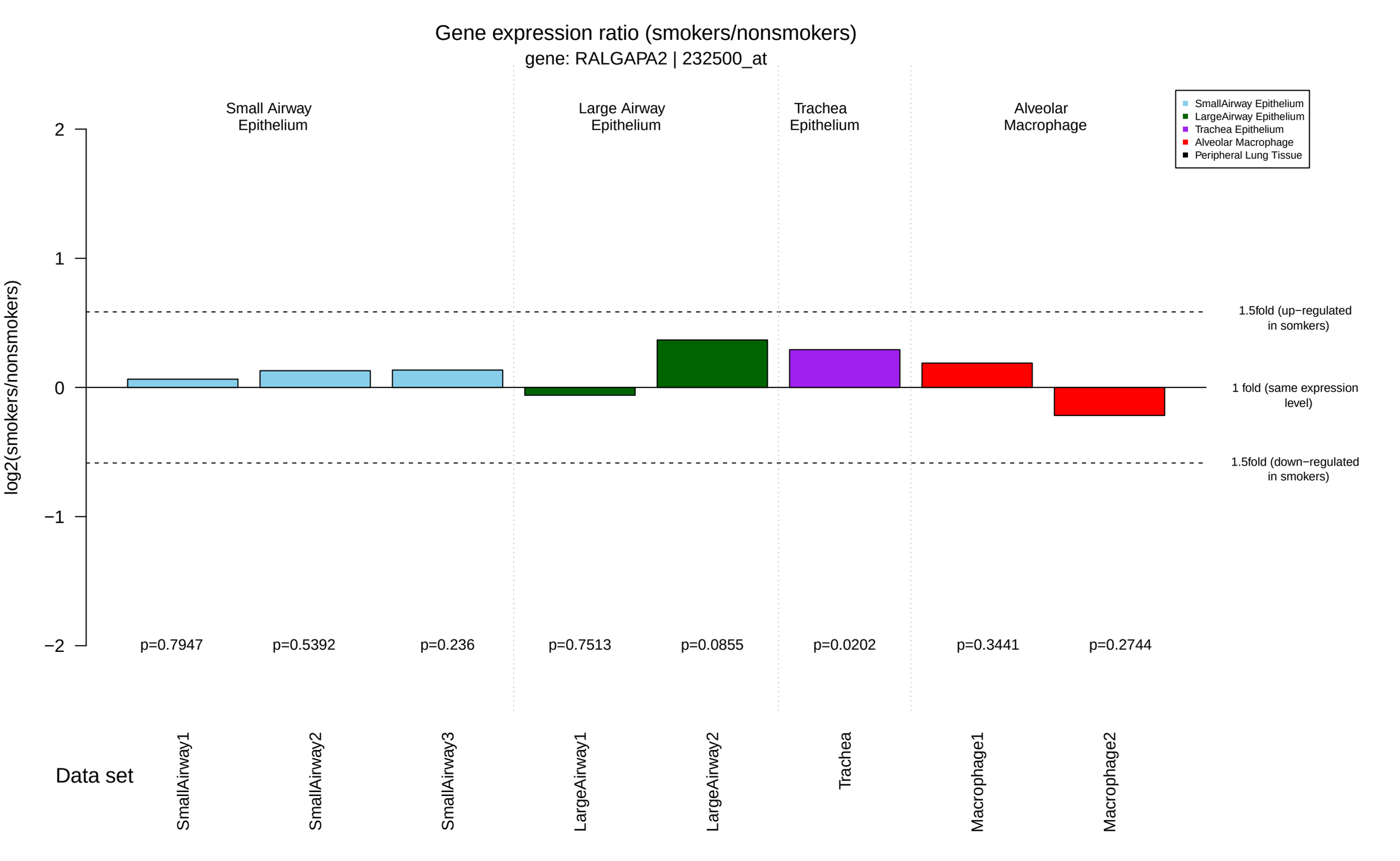
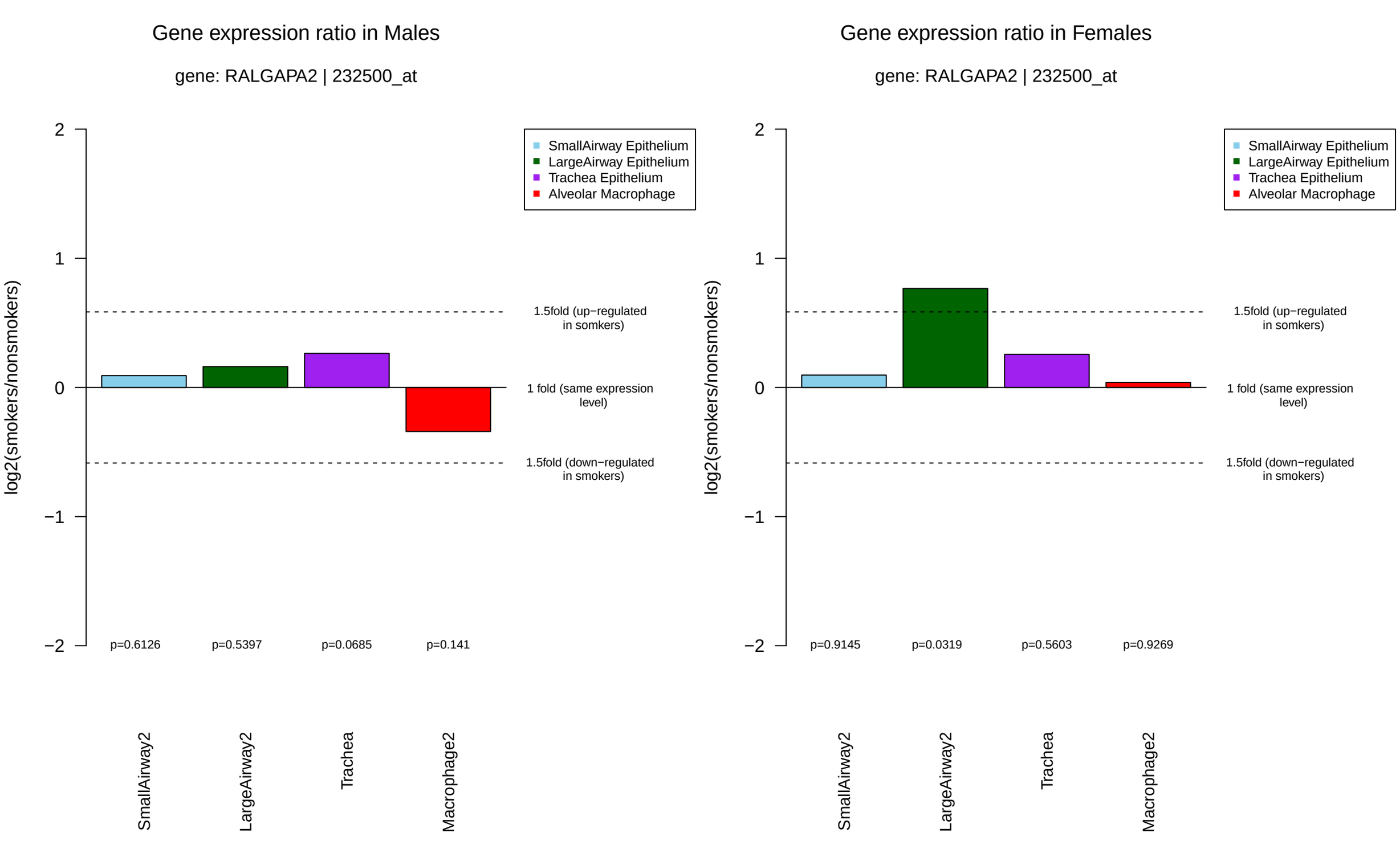
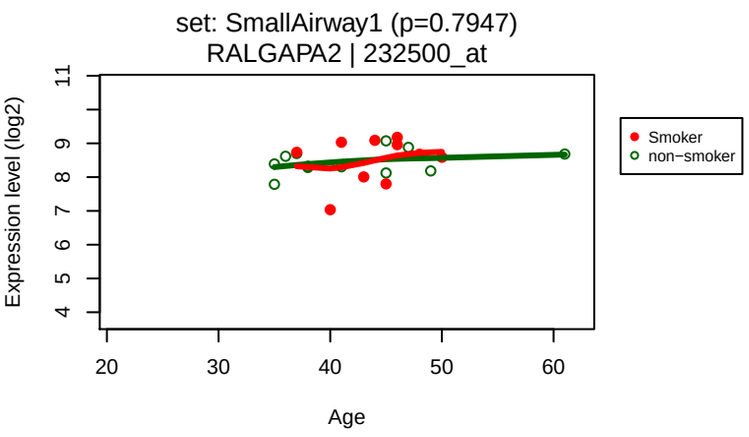
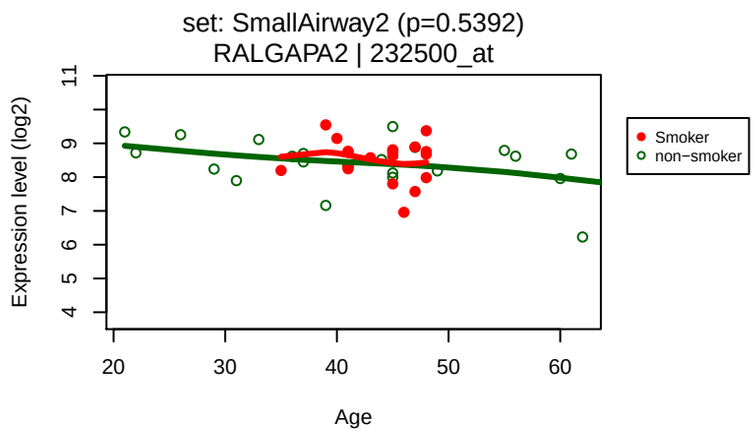
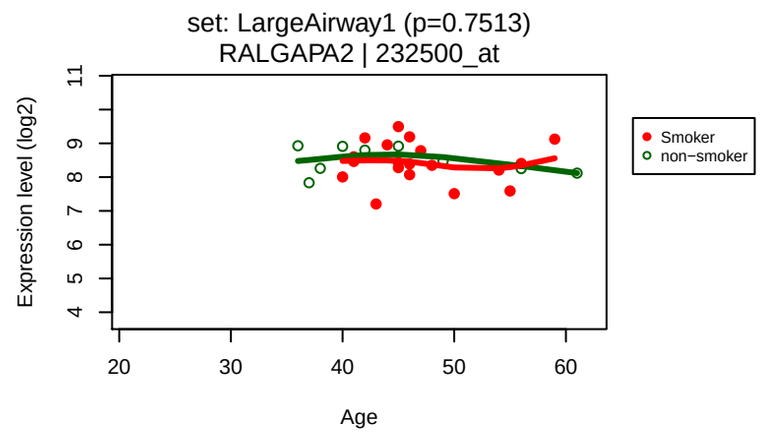

Gene expression of smokers and nonsmokers
gene: RALGAPA2 | 232500_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7947
p=0.5392
p=0.236
p=0.7513
p=0.0855
p=0.0202
p=0.3441
p=0.2744
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

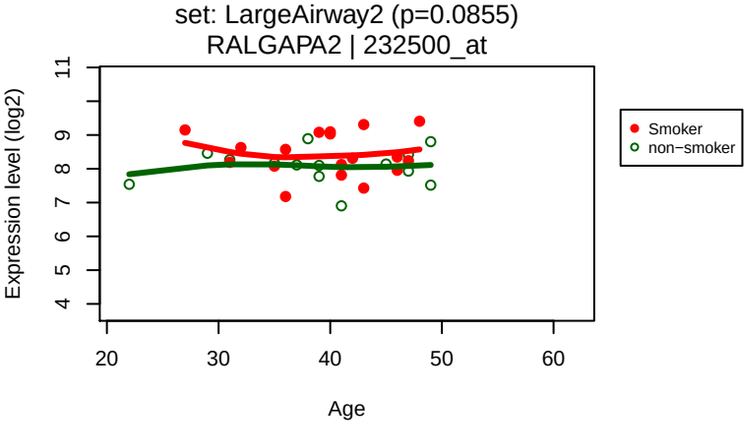

Gene expression ratio (smokers/nonsmokers)
gene: RALGAPA2 | 232500_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7947
p=0.5392
p=0.236
p=0.7513
p=0.0855
p=0.0202
p=0.3441
p=0.2744
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

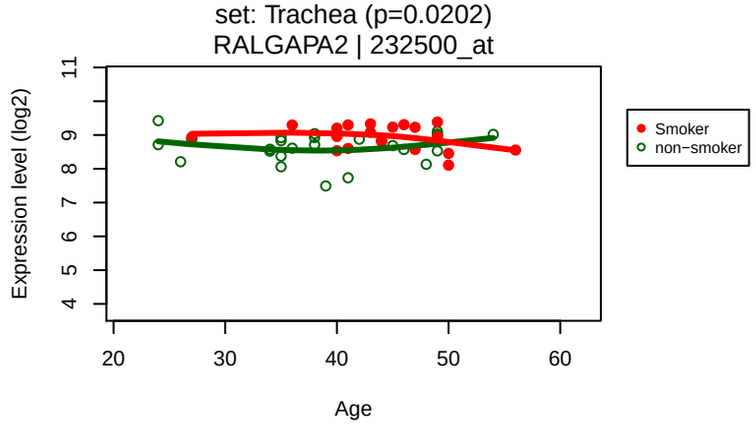

Gene expression ratio in Males
gene: RALGAPA2 | 232500_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6126
p=0.5397
p=0.0685
p=0.141
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RALGAPA2 | 232500_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9145
p=0.0319
p=0.5603
p=0.9269
SmallAirway2
LargeAirway2
Trachea
Macrophage2

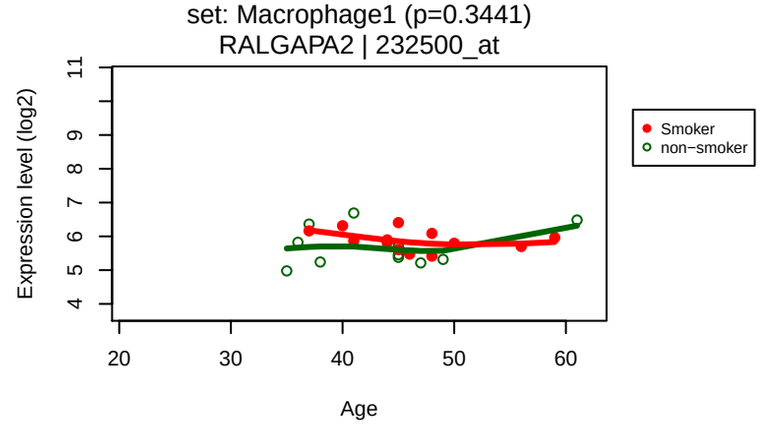

set: SmallAirway1 (p=0.7947)
RALGAPA2 | 232500_at
Expression level (log2)
Age
Smoker
non−smoker

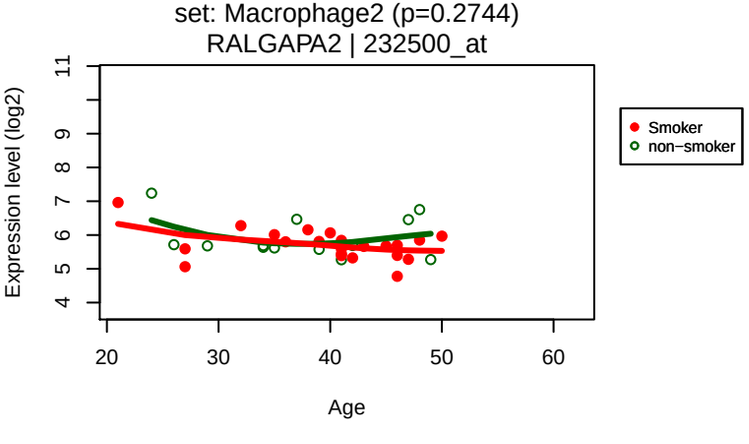

set: SmallAirway2 (p=0.5392)
RALGAPA2 | 232500_at
Expression level (log2)
Age
Smoker
non−smoker

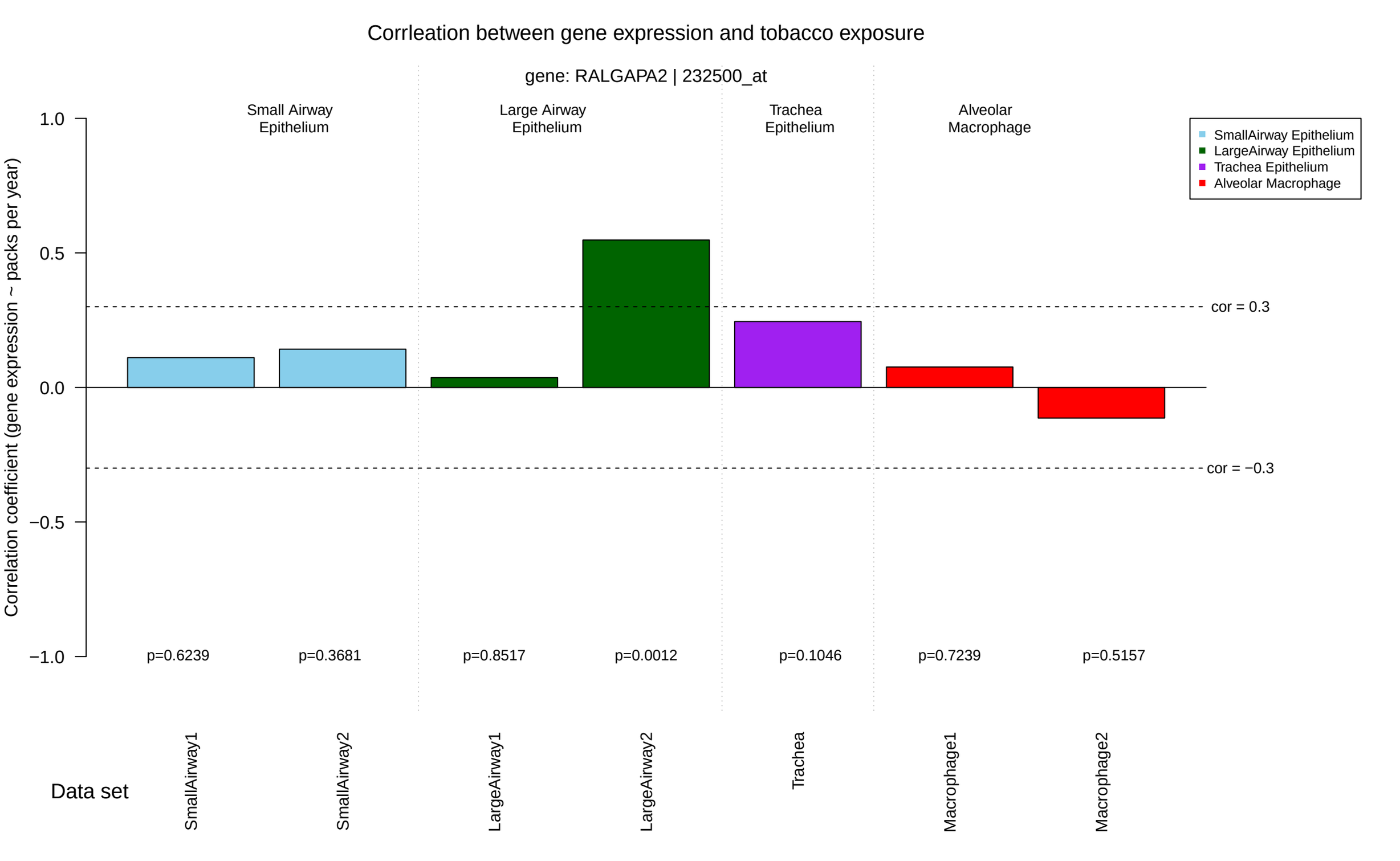

set: LargeAirway1 (p=0.7513)
RALGAPA2 | 232500_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0855)
RALGAPA2 | 232500_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0202)
RALGAPA2 | 232500_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3441)
RALGAPA2 | 232500_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2744)
RALGAPA2 | 232500_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RALGAPA2 | 232500_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6239
p=0.3681
p=0.8517
p=0.0012
p=0.1046
p=0.7239
p=0.5157
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2